

IPL Crunch '26 — Decoding Winning Patterns in the IPL Using Data Analytics

Introduction

IPL conversations are full of strong opinions:

- “Winning the toss is a huge advantage.”
- “Death overs decide T20 matches.”
- “Big totals are defended through economical bowling.”

This project tests those beliefs using real IPL ball-by-ball data.

Using Python, Pandas, and Tableau, I analyzed IPL matches across multiple seasons to uncover patterns in:

- Toss impact
- Match-winning phases
- Batting and bowling strategies
- Pressure handling
- Top performers
- Hidden insights that challenge common cricket assumptions

The goal was not just to create charts, but to explain *why teams actually win IPL matches*.

Dataset Overview

The dataset contains ball-by-ball IPL match data where every row represents one delivery bowled in an IPL match.

The data includes:

- Match details
- Toss decisions
- Batting and bowling events
- Runs scored
- Wickets taken
- Match winners

Key columns used:

- match_id
- season
- venue
- toss_winner
- toss_decision
- winner
- batting_team
- batter
- bowler
- runs_total
- wicket_kind

Before analysis:

- Missing values were handled
- Team naming inconsistencies were cleaned
- “Royal Challengers Bangalore” was standardized to “Royal Challengers Bengaluru”

This ensured accurate aggregation and comparisons.

Analysis 1 — Does Winning The Toss Actually Matter?

Objective

A common belief in cricket is that winning the toss gives teams a major advantage.

This analysis tested whether toss winners actually win significantly more IPL matches.



Impact of Toss on IPL Match Outcomes

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Methodology

For every match:

- Toss winner was identified
- Match winner was identified
- Win percentages were calculated for:
 - Toss winners
 - Toss losers

Results

Outcome	Win Rate
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Toss 50.15%
Winners

Toss Losers 49.85%

Difference: only 0.30%

Insight

The toss has almost no impact on overall IPL results.

Even though commentators often emphasize toss decisions, the data shows that toss winners and toss losers win almost the same number of matches.

This suggests:

- Team performance matters far more than toss advantage
 - Toss may influence strategy at specific venues, but not league-wide outcomes
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Analysis 2 — Where Winning Teams Pull Ahead

Objective

The innings was divided into three phases:

- Powerplay (Overs 1–6)
- Middle Overs (Overs 7–15)
- Death Overs (Overs 16–20)

The goal was to compare scoring patterns between winning and losing teams.

Results

Phase	Winning Teams	Losing Teams	Difference
Powerplay	58.0	50.2	+7.8
Middle Overs	80.9	73.4	+7.5
Death Overs	49.9	45.9	+4.0

Insight

Winning teams separate themselves early and during the middle overs — not at the end.

The largest scoring gaps appear during:

- Powerplay
- Middle Overs

Death Overs showed the smallest difference between winners and losers.

This suggests:

- Strong starts and middle-over control matter more than late acceleration
 - Teams that dominate earlier phases usually enter death overs already ahead
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Analysis 3 — Which Phase Actually Decides The Match?

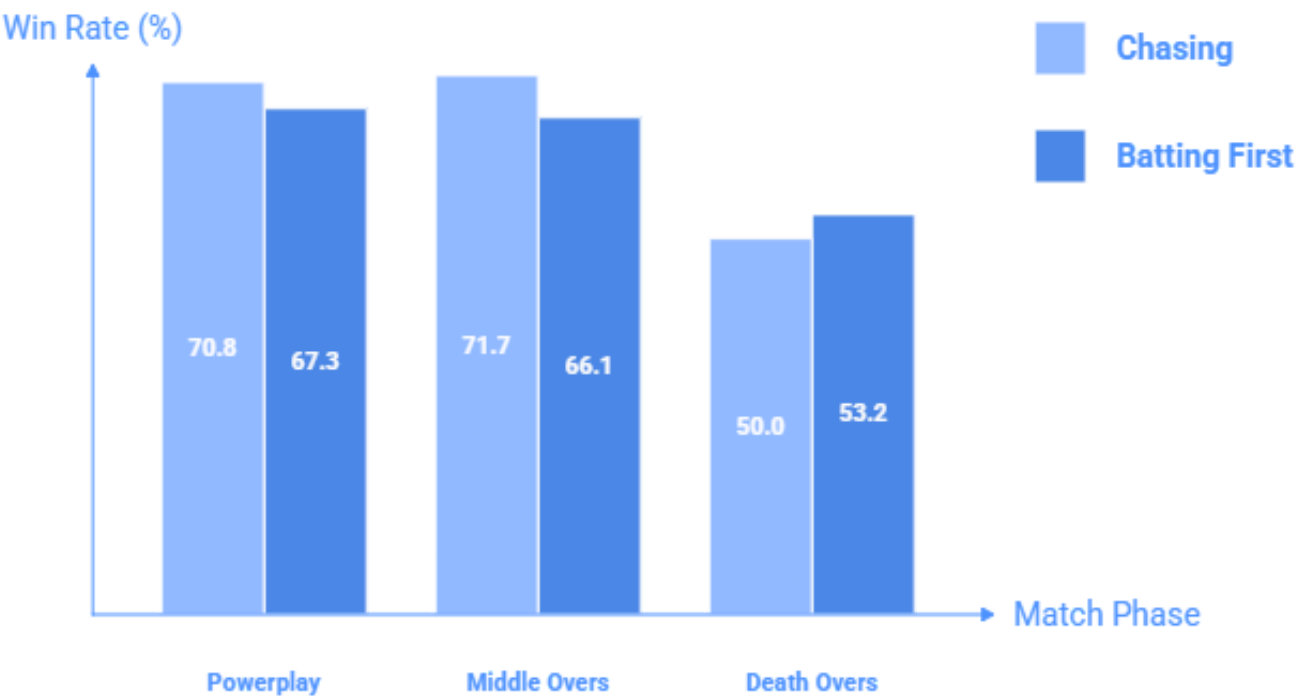
Objective

This analysis measured win percentage when teams dominate a specific phase.

A team was considered to dominate a phase if it scored more runs than the opponent in that phase.

The analysis was separated into:

- Teams batting first
- Teams chasing



Win Rate by Phase Domination for Batting First and Chasing Teams

Made with Napkin

Batting First Results

Phase Dominated	Win Rate
Powerplay	67.3%
Middle Overs	66.1%

Death Overs 53.2%

Chasing Results

Phase Dominated	Win Rate
Powerplay	70.8%
Middle Overs	71.7%
Death Overs	50.0%

Biggest Surprise Finding

Death Overs are almost a coin flip.

Despite being considered the most important phase in T20 cricket:

- Dominating Death Overs while chasing gives only a 50.0% win rate.
- For teams batting first, the advantage rises to just 53.2%.

This means Death Overs barely influence match outcomes compared to earlier phases.

Why This Matters

This was the strongest insight discovered in the project because it directly challenges conventional cricket thinking.

The data suggests:

- IPL matches are usually decided before the death overs begin.
- Teams that dominate Powerplay and Middle Overs build winning positions early.
- Death Overs often only amplify an already established trend.

This was the most surprising pattern uncovered in the analysis.

Analysis 4 — Bowling Strategy Changes Based On Total Defended

Objective

This analysis studied whether:

- Taking wickets
or
- Bowling economically

is more important while defending totals.

The analysis was split into:

- Defending 180+
 - Defending Below 180
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Bowling Strategy vs. Total Defended



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Results

Defending 180+

Bowling Type	Successful Defense Rate
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Wickets + Economical	86.9%
Wickets + Expensive	70.7%
No Wickets + Economical	74.6%
No Wickets + Expensive	52.1%

Defending Below 180

Bowling Type	Successful Defense Rate
Wickets + Economical	47.9%
Wickets + Expensive	48.7%
No Wickets + Economical	23.5%
No Wickets + Expensive	11.2%

Insight

Bowling strategy completely changes depending on the total being defended.

When Defending 180+

Economy becomes highly valuable because scoreboard pressure naturally increases.

Even without wickets:

- Economical bowling alone wins 74.6% of matches.
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When Defending Below 180

Wickets become everything.

Being economical without wickets wins only 23.5% of matches.

This suggests:

- Low totals require attacking bowling
- High totals allow teams to defend through scoreboard pressure and control

Analysis 5 — Dot Ball Pressure

Another hidden pattern emerged while studying pressure situations.

The data showed:

- Pressure situations create more dot balls rather than immediate wickets.
- Winning teams remain calmer during high-pressure phases.
- Sustained dot-ball pressure gradually forces mistakes instead of causing instant collapses.

This highlights the importance of momentum control in T20 cricket.

Top Performers Across Seasons

Season	Top Batter	Runs	Top Bowler	Wickets
2022	JC Buttler	863	YS Chahal	29
2023	Shubman Gill	890	MM Sharma	31
2024	V Kohli	741	HV Patel	30
2025	B Sai Sudharsan	759	M Prasidh Krishna	26
2026	H Klaasen	494	B Kumar	18

Visualization Summary

The dashboard includes:

- Toss impact analysis
- Phase comparison charts
- Match-winning phase analysis
- Bowling strategy analysis
- Top performers table
- Pressure pattern insights

The visualizations were designed to:

- maximize readability
 - highlight comparisons clearly
 - support storytelling
 - communicate insights quickly
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Tools Used

- Python
 - Pandas
 - NumPy
 - Tableau
 - Jupyter Notebook
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Final Conclusion

This project used IPL ball-by-ball data to test popular cricket assumptions using analytics.

The analysis revealed that:

- Winning the toss provides almost no real advantage.
- Powerplay and Middle Overs are far more predictive of winning than Death Overs.
- Bowling strategy changes completely depending on the total being defended.
- Pressure in T20 cricket is often created through sustained dot balls rather than sudden wickets.

Most importantly, the project uncovered a surprising insight:

Death Overs are statistically close to a coin flip.

Despite their reputation, they are the least predictive phase of the innings.

The data suggests that successful IPL teams usually build control much earlier — through strong starts, middle-over dominance, and sustained pressure.

This project demonstrates how data analytics can challenge assumptions and uncover deeper strategic patterns in modern cricket.